Project: The 'Green-Ops' Carbon Tracker

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Course: BCA

Reinforcement Learning for Sustainable Cloud Computing

Problem Statement

Build a complete, real-world OpenEnv environment that an AI agent can learn from through the standard **step() / reset() / state()** API.

KEY REQUIREMENTS AT A GLANCE

- Must simulate a real-world task (not games or toys)
- Implement full OpenEnv spec: typed models, step()/reset()/state(), openenv.yaml
- Minimum 3 tasks with agent graders (easy → medium → hard, scores/reward 0.0–1.0)
- Meaningful reward function with partial progress signals
- Baseline inference script with reproducible scores
- Deploy to Hugging Face Spaces + working Dockerfile
- README with environment description, action/observation spaces, setup instructions

Introduction

- Cloud computing consumes large amounts of electricity.
- Many data centers rely on fossil fuels which increase carbon emissions.
- Sustainable IT aims to reduce environmental impact using intelligent systems.
- This project uses Reinforcement Learning to optimize server usage and reduce carbon emissions.

Project Overview

- The Green-Ops Carbon Tracker simulates a cloud infrastructure environment.
- An AI agent observes server states and chooses actions.
- Goal: Reduce carbon emissions while maintaining server performance.
- The system uses Reinforcement Learning decision making.

Technologies Used

- Programming Language: Python
- Framework: OpenEnv
- AI Model: Hugging Face LLM
- Concept: Reinforcement Learning
- Simulation: Cloud server energy optimization

Environment State

- The environment stores data in JSON format.
- Each server has:
- - Server ID
- - CPU Load
- - Power Source (Coal, Solar, Wind)
- Example:
- {server_id:1, cpu_load:80, power:Coal}

Tasks in the Environment

- The agent must complete three levels of tasks.

- Easy Task: Check carbon score.
- Medium Task: Move high CPU load from coal server to solar server.
- Hard Task: Schedule backup only when wind power is at 100%.

Carbon Score Logic

- Carbon score represents environmental impact.

- Coal Power -> High emissions
- Solar Power -> Low emissions
- Wind Power -> Very low emissions

- The agent tries to reduce the overall carbon score.

Reinforcement Learning Process

1. Agent observes the environment state.
2. Agent selects an action.
3. Environment returns reward.
4. Agent learns which actions reduce carbon emissions.
5. The process repeats until optimal decisions are learned.

Benefits of the Project

- Encourages sustainable cloud computing.
- Demonstrates AI decision making.
- Shows real-world application of reinforcement learning.
- Helps reduce energy waste in data centers.

Conclusion

- The Green-Ops Carbon Tracker demonstrates how AI can support Sustainable IT.
- Reinforcement Learning helps optimize server workloads.
- This approach can help reduce carbon emissions in future cloud infrastructures.

Thank You